Cryptographic Privacy. Privacy in Big Data

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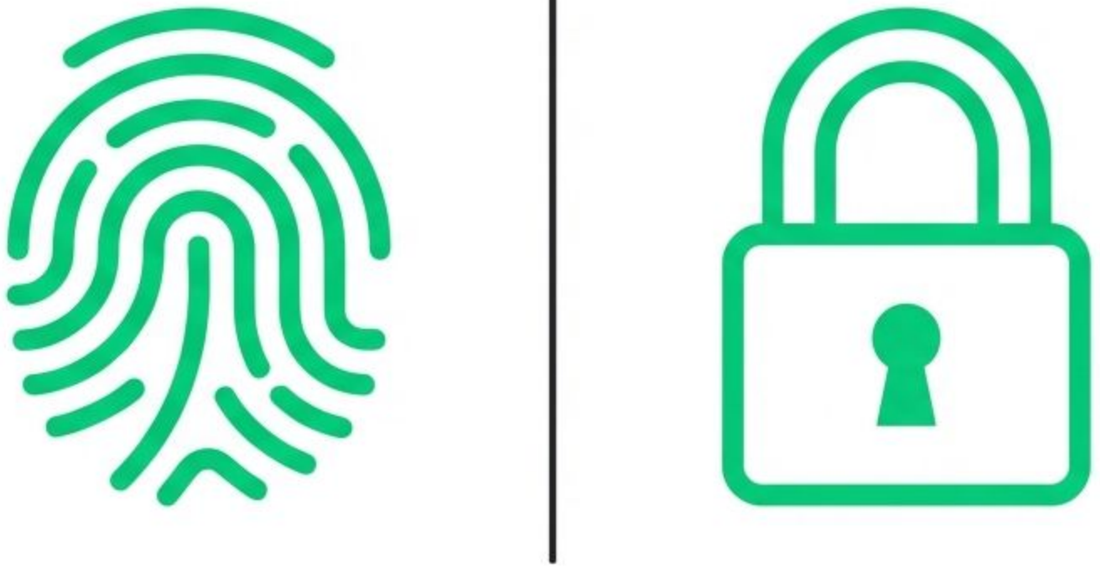
AGENDA:

1. Cryptographic Foundations:
   - Zero knowledge proof
   - The Computation Problem.
2. The Big Data Privacy Challenge
   - Structured and unstructured data

Cryptographic Foundations.

Zero knowledge proof

The Modern Privacy Dilemma



How can we benefit from new technologies without surrendering our sensitive data?

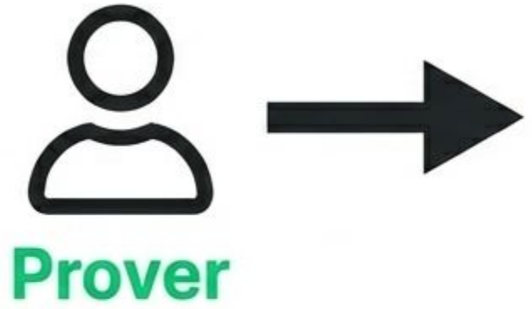
We Want Both.



We want to embrace powerful new services. But we also need to protect our privacy.

Privacy Enhancing Technologies (PETs) aim to solve this.

Introducing the Zero-Knowledge Proof (ZKP)



Prover

A protocol which lets a **Prover** prove a **Statement** about a secret is true, without revealing the secret itself.

Intuition #1: The Two Pens

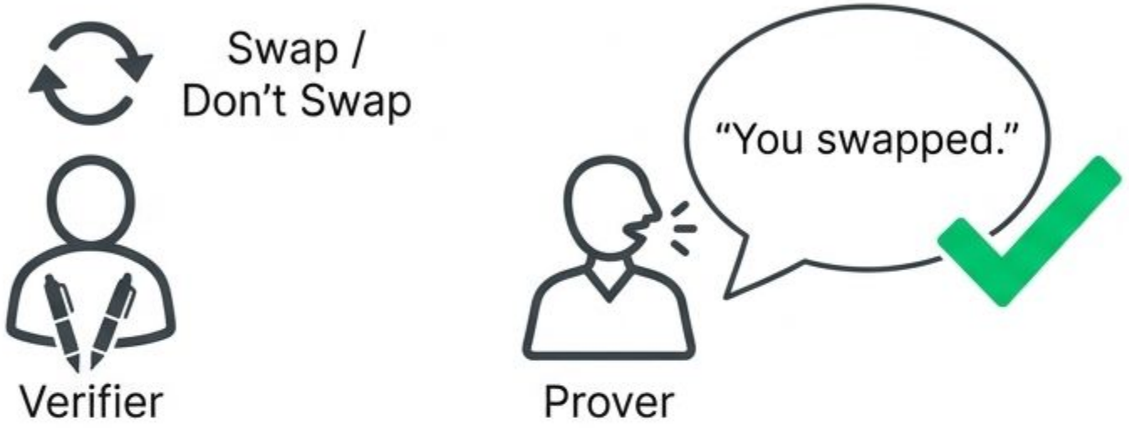


The Prover: Claims they can see a difference between the red and green pens.

The Verifier (You): Is colorblind and sees two identical pens. You don't believe the Prover and think they might be lying or just guessing.

The Challenge: How can the Prover convince you the pens are truly different without ever telling you which color is which?

The Proof is in the Repetition.



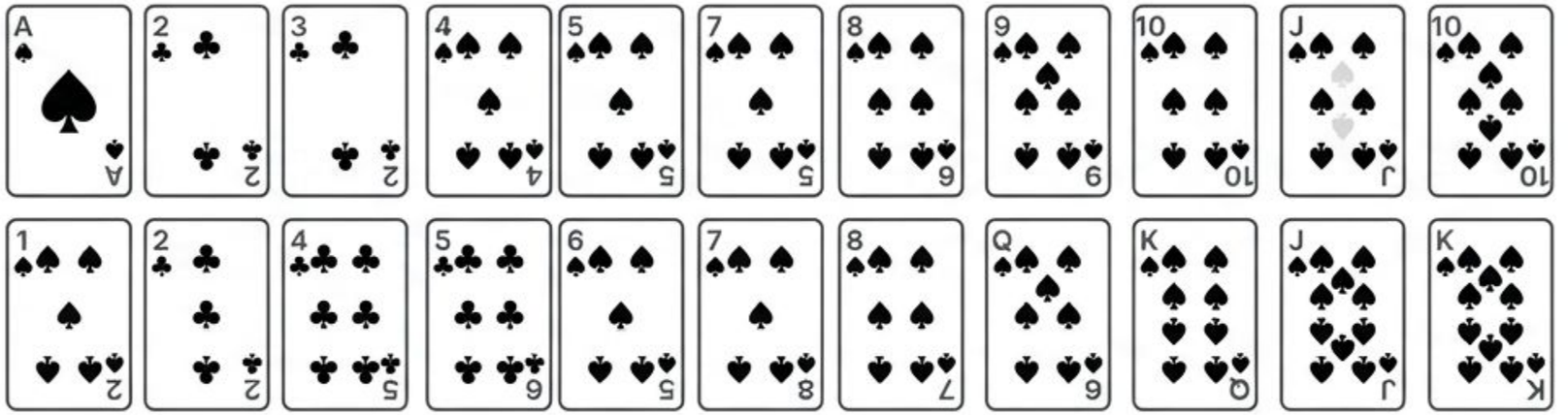
$$\frac{1}{2} \rightarrow \frac{1}{4} \rightarrow \frac{1}{8}$$

- You hide the pens and randomly choose to swap them or not.
- I tell you whether you swapped.

- **Round 1: 50% chance I guessed.**
- **Round 2: 25% chance I guessed.**
- **Round 3: 12.5% chance I guessed.**
- **We repeat until you are convinced I’m not cheating.**

Intuition #2: The Red Card





Scenario Text:

- I have one secret card from a 52-card deck.
- My Goal: Prove to you the secret card is red, without revealing its number or suit.

The Proof: I show you all 26 black cards from the rest of the deck. If they are all accounted for, my secret card ***must*** be red.

The Three Pillars of a Zero-Knowledge Proof



Correctness

If the prover is honest and knows the secret, the proof will work.



Soundness

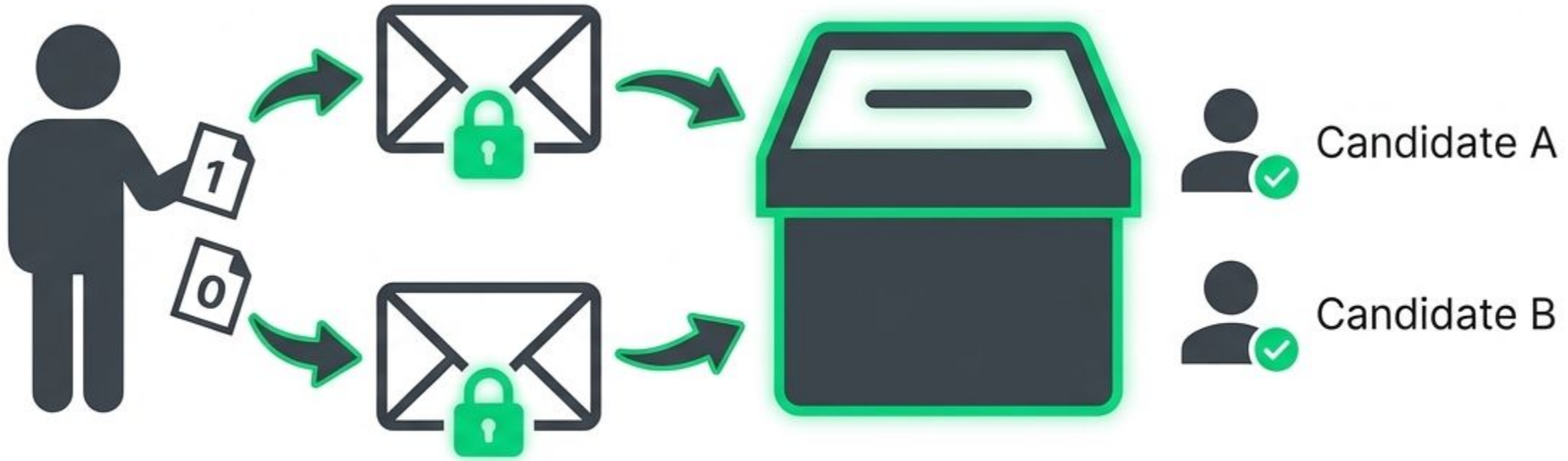
A dishonest prover who does *not* know the secret cannot fake the proof.



Zero-Knowledge

The verifier learns nothing beyond the fact that the statement is true. No side information is leaked.

Application: Building a Secure E-Voting System



The Setup:

- Votes are encrypted (placed in 'envelopes').
- **Goal:** Aggregate the results without decrypting individual votes.
- **Problem:** A naive system can be easily cheated by voters.

Three Ways a Voter Can Cheat



Problem 1: Invalid Sum

A voter submits **(Vote: 1, Vote: 1)**. They voted for both candidates, not just one.



Problem 2: Invalid Values

A voter submits **(Vote: +2, Vote: -1)**. The sum is 1, but the votes themselves are malicious and invalid.



Problem 3: The Copy-Paste Attack

A voter copies another person's encrypted vote without knowing what's inside, undermining the principle of an informed vote.

ZKP Solution #1: Prove the Sum is One

$$VOTE_A + VOTE_B = 1$$



The Statement: "I am proving that the sum of the encrypted values in my envelopes is exactly 1."

The Secret: The actual votes (0 and 1).

The Result: This ensures each person votes for exactly one candidate.

ZKP Solution #2: Prove the Votes are Binary

$$VOTE \in \{0, 1\}$$



The Statement: “I am proving that each encrypted value is either a 0 or a 1.”

The Secret: The actual votes (`0` and `1`).

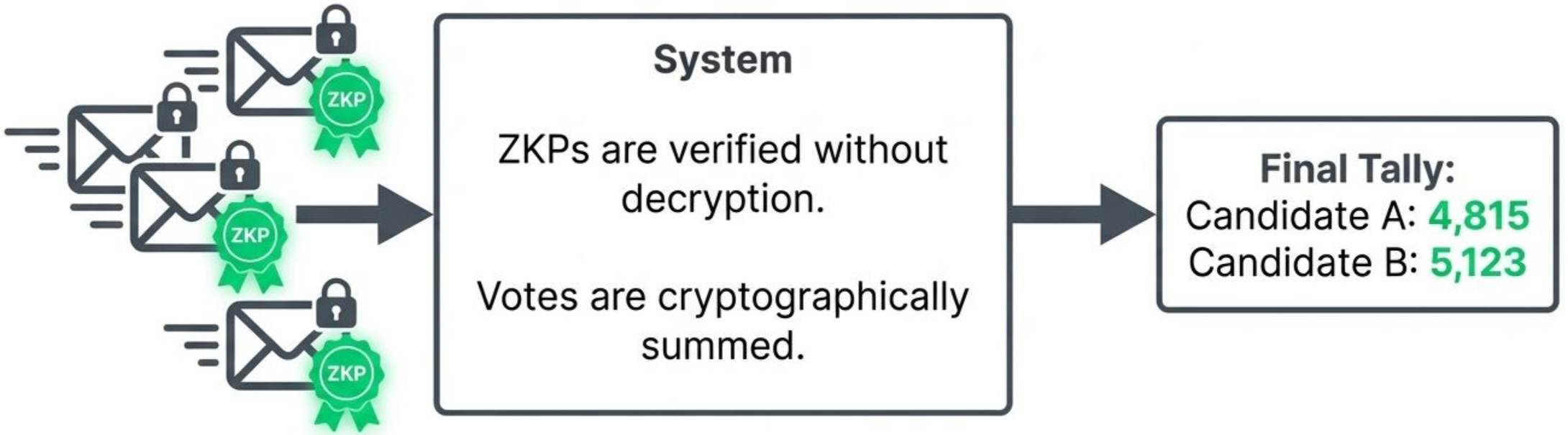
The Result: This mitigates the `+2 / -1` attack and ensures only valid votes are cast.

ZKP Solution #3: Prove You Know Your Vote



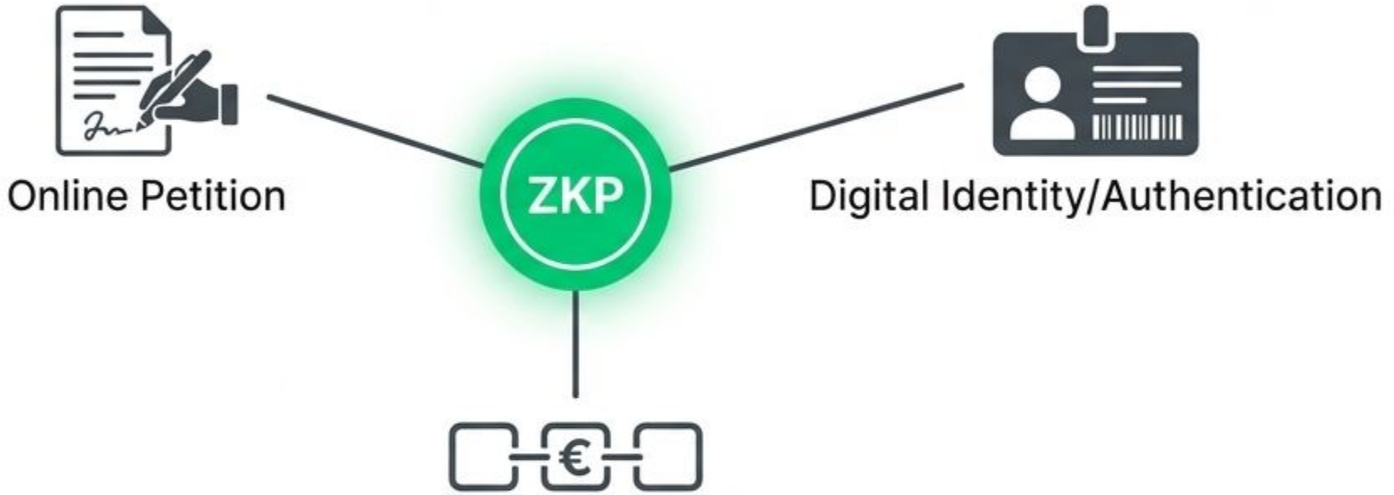
- The Statement: “I am proving that I know the secret contents of the envelopes I am submitting.”
- The Secret: The decryption key for the vote, or the vote itself.
- The Result: This prevents a voter from blindly submitting someone else’s encrypted vote.

Verification Without Revelation



Key Point: The ZKP protocol works *with* the encryption scheme. No one can decrypt individual votes. The system only ever decrypts the final, aggregated sum after all votes are cast and verified.

More Than Just Voting



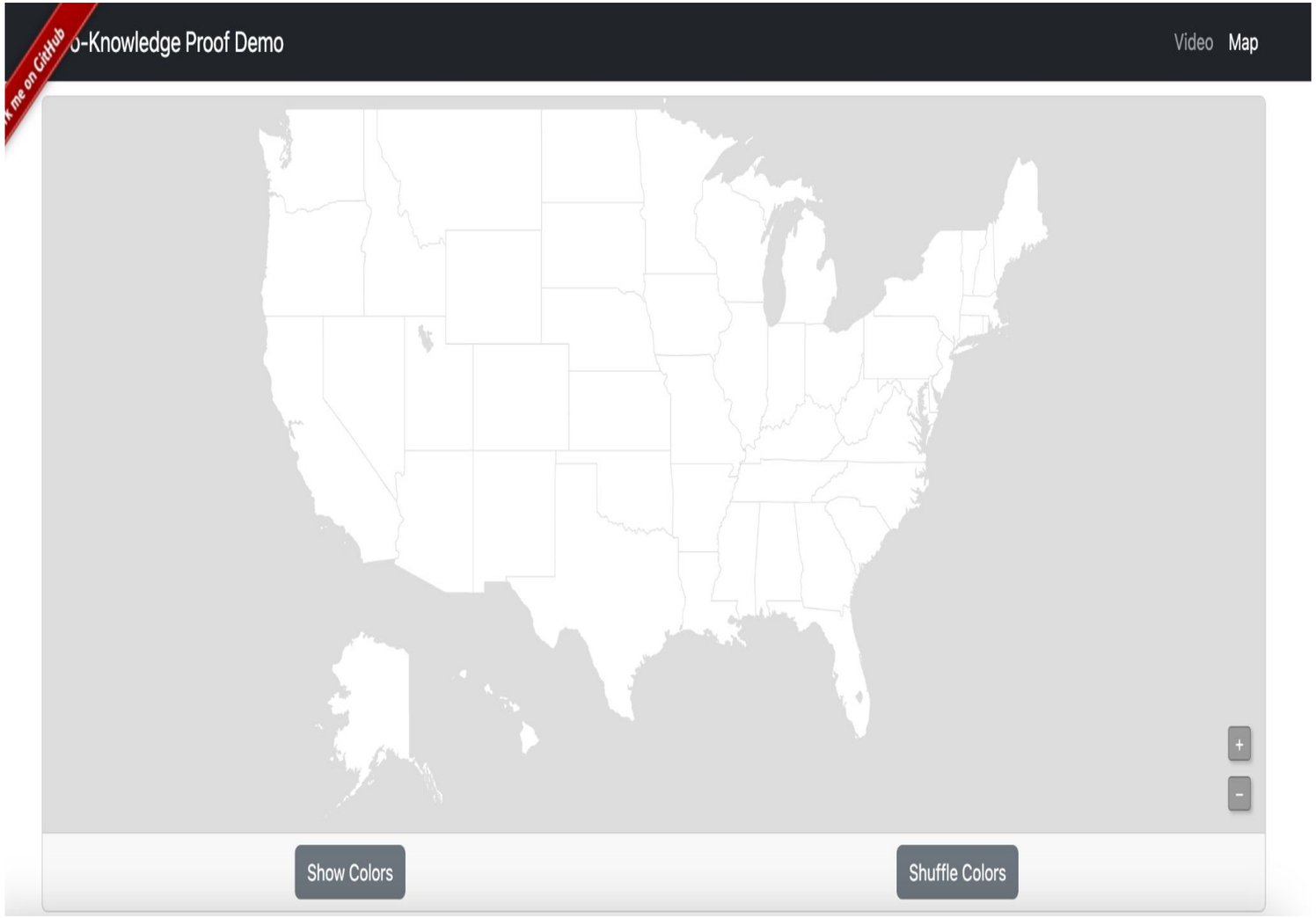
Confidential Digital Payments

The same principles can be applied to:

- Secure and private online petitions.
- Anonymous authentication systems.
- Confidential transactions in digital currencies.
- And many other emerging use cases.



https://andersbrownworth.com/zero-knowledge-proof-demo/



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The Computation Problem.

Secure Multiparty Computation & The "Data-in-Use" Gap

We’ve Mastered Two States of Data Protection



Data at Rest

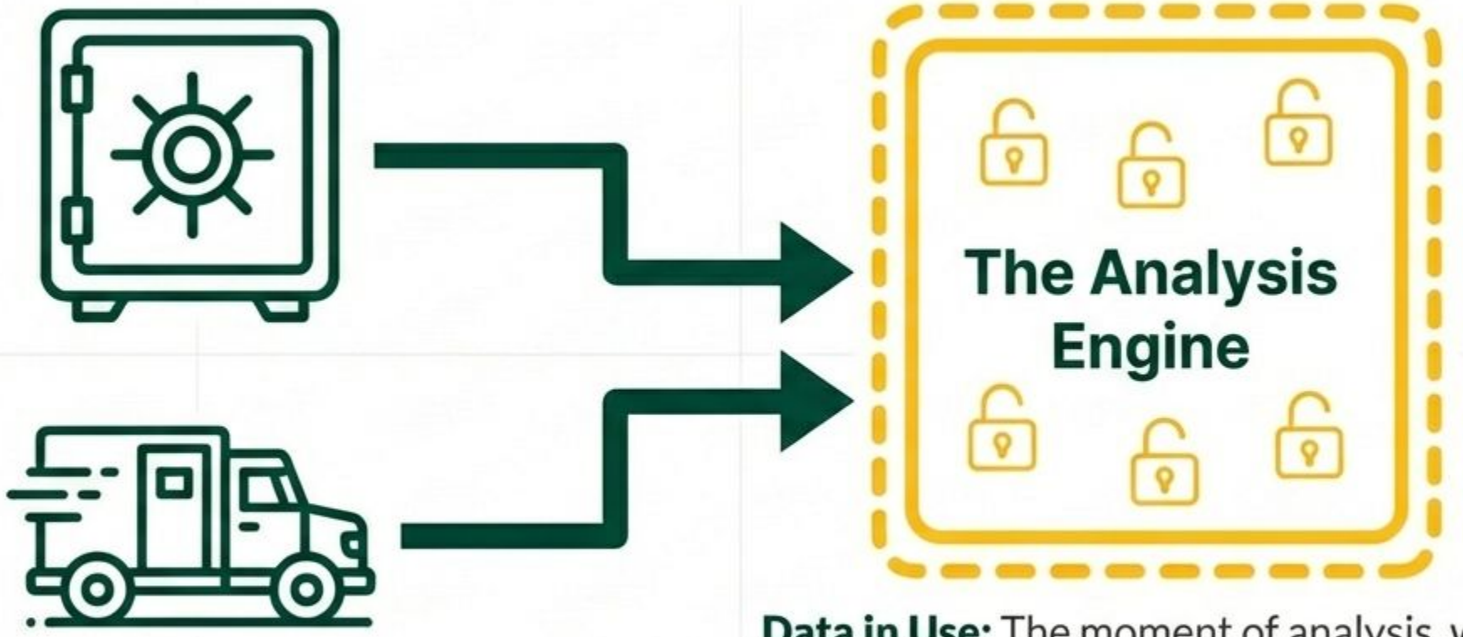
Protected by encryption in storage (e.g., disk encryption).



Data in Transit

Protected by encryption during transfer (e.g., TLS/SSL).

But What Happens When Data is Actually Used?



Data in Use: The moment of analysis, where data must be decrypted to be processed, creating a critical point of vulnerability.

How can we perform analysis while keeping sensitive data protected?

Three Emerging Cryptographic Solutions

A new class of cryptographic techniques aims to protect sensitive data during the analysis phase. While they have overlapping functions, they are distinct approaches that can often be combined to enhance security.



Secure Multiparty Computation (MPC)

A cryptographic protocol that allows multiple parties to jointly compute a function over their inputs while keeping those inputs private.



Confidential Computing

A security technology that isolates sensitive data in a protected CPU enclave during processing, preventing unauthorized access.



(Fully) Homomorphic Encryption

A form of encryption that permits computations on encrypted data, generating an encrypted result that matches the result of operations on the plaintext.

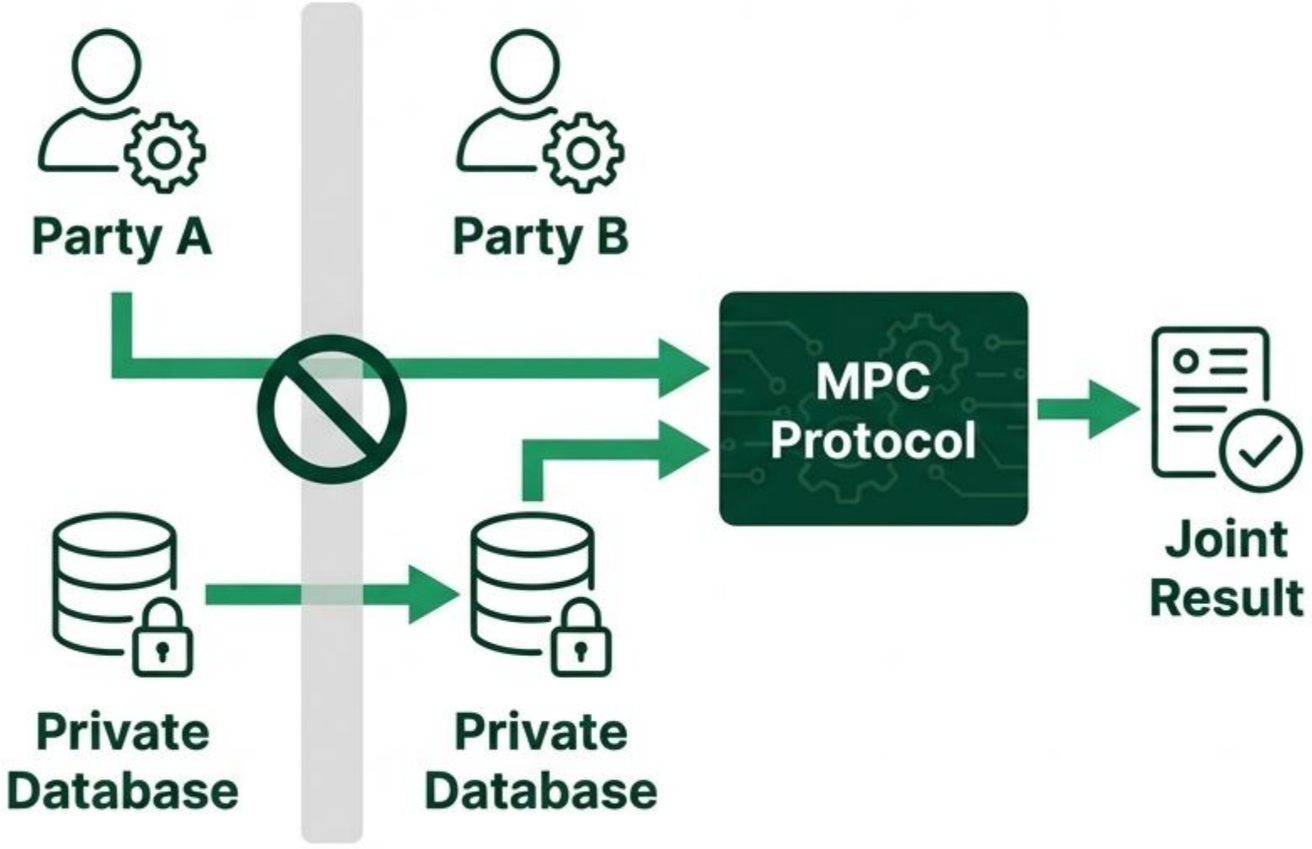


I. Secure Multiparty Computation (MPC)

A set of techniques allowing multiple parties to jointly perform analyses on distributed datasets without revealing the underlying data to each other.

"Analyze a shared database you can't see."

The Principle: Joint Computation, Private Data



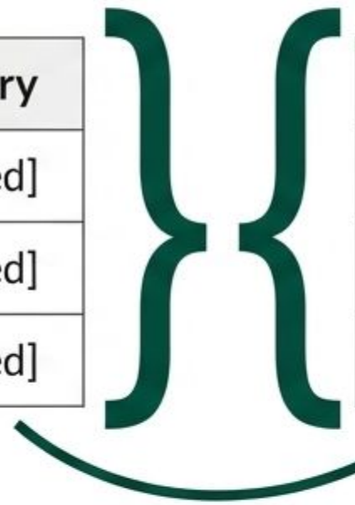
- No raw data is shared between parties.
- Computations are guaranteed to perform correctly.
- Provides control over who receives the final computation result.
- A powerful implementation of “Privacy by Design.”

When to Use MPC

Primary Use Case: Ideal for linking "**vertically partitioned**" datasets.

Hospital Data

Person ID	Medical History
1	[Data Encrypted]
2	[Data Encrypted]
3	[Data Encrypted]



Insurance Data

Person ID	Insurance Claims
1	[Data Encrypted]
2	[Data Encrypted]
3	[Data Encrypted]

This is when different organizations hold different *types* of information about the *same* individuals and need to link these sources for analysis.

MPC in Action: Healthcare Analytics



Scenario: A medical insurance company and a hospital needed to determine the effectiveness of a diabetes app.

MPC Application: They calculated the average medical costs for patient groups based on app usage.

The Result: The analysis was completed **without revealing any individual patient information** between the insurance company and the hospital.

Implications & Research Considerations



1. Joint Agreement is Non-Negotiable: All parties must agree agree in advance on the specific analysis to be performed and what results will be revealed.



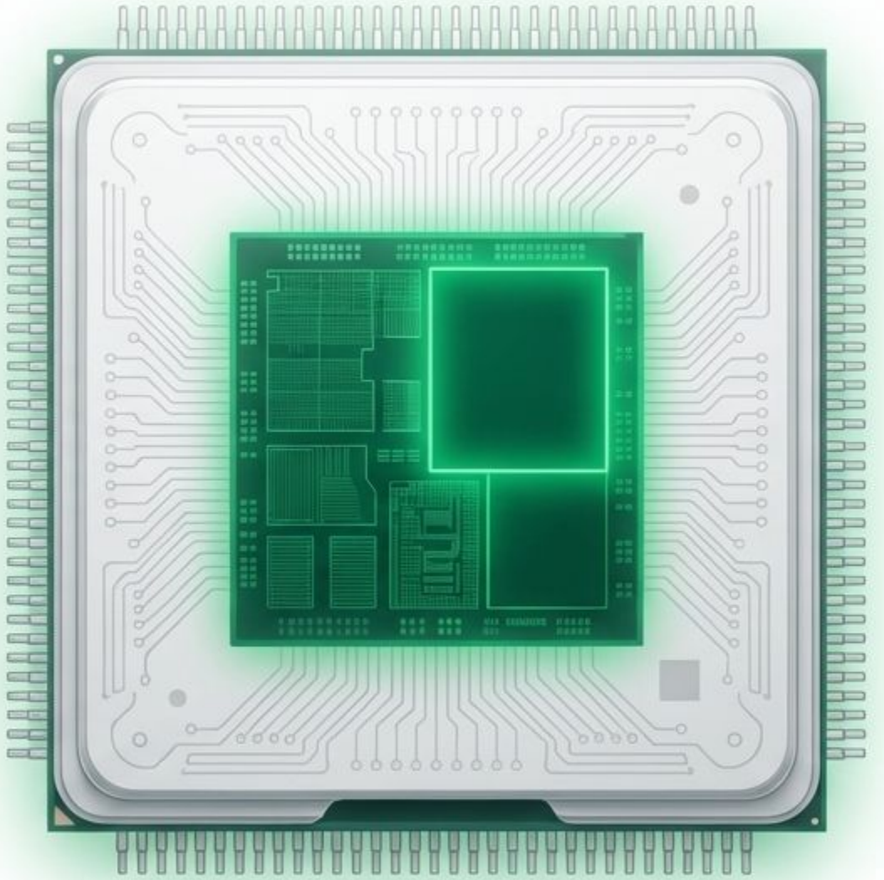
2. No One-Size-Fits-All Solution: Different use cases require different, often custom, implementations of MPC.



3. Requires Additional Resources: Significant computational overhead is needed to generate random secrets and distribute data among the parties.

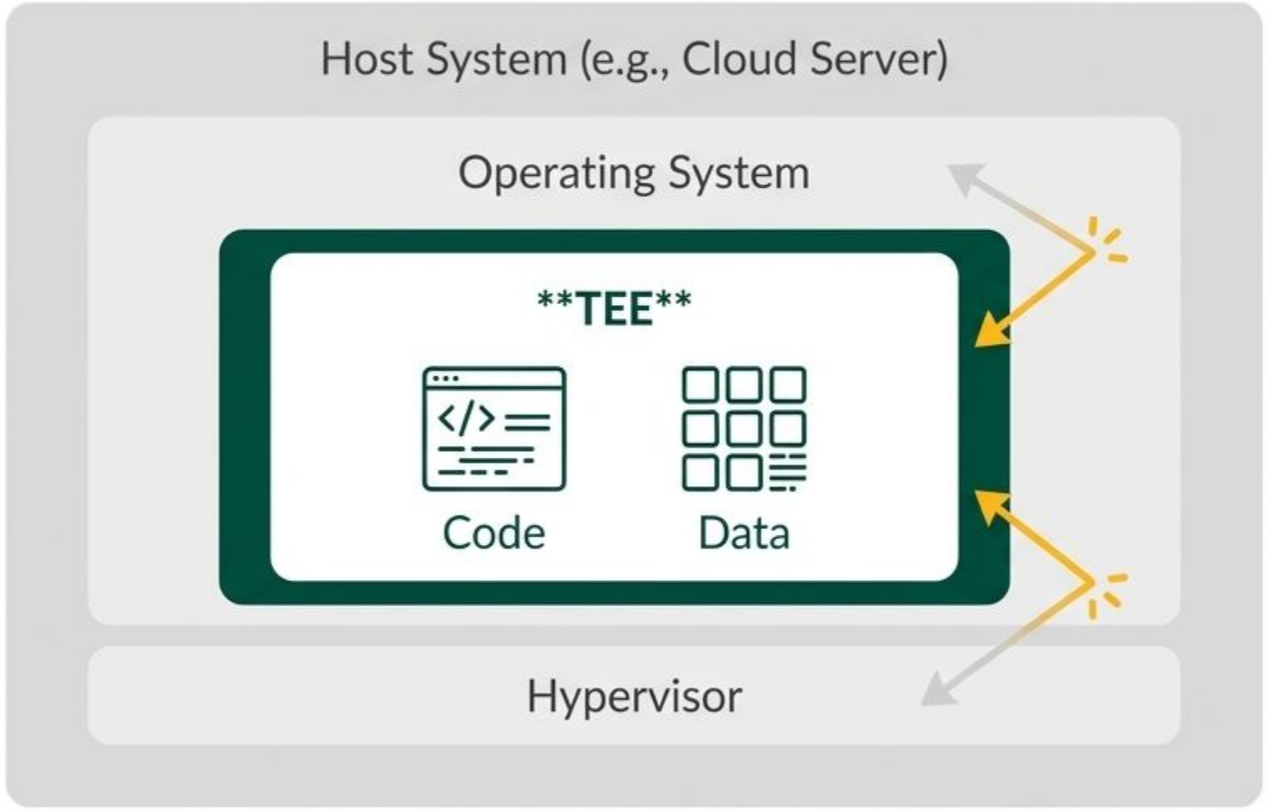
II. Confidential Computing

A technique that protects data in use by isolating it within a hardware-based Trusted Execution Environment (TEE).



“A secure vault inside the processor.”

How It Works: Creating an Isolated Environment



Key Protections:

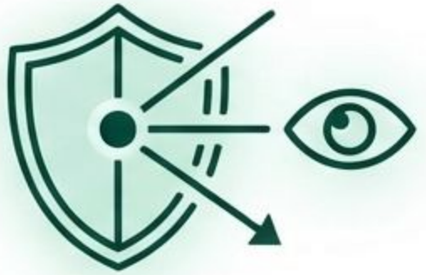
- Data Confidentiality: Prevents data from being viewed.
- Data & Code Integrity: Prevents data and the running code from being modified or deleted.

Total Isolation: Data and code inside the TEE are invisible even to the operating system, cloud provider, and any virtual machines.

When to Use Confidential Computing



Use Case 1: Analyzing sensitive data on an **untrusted cloud platform or third-party infrastructure.**

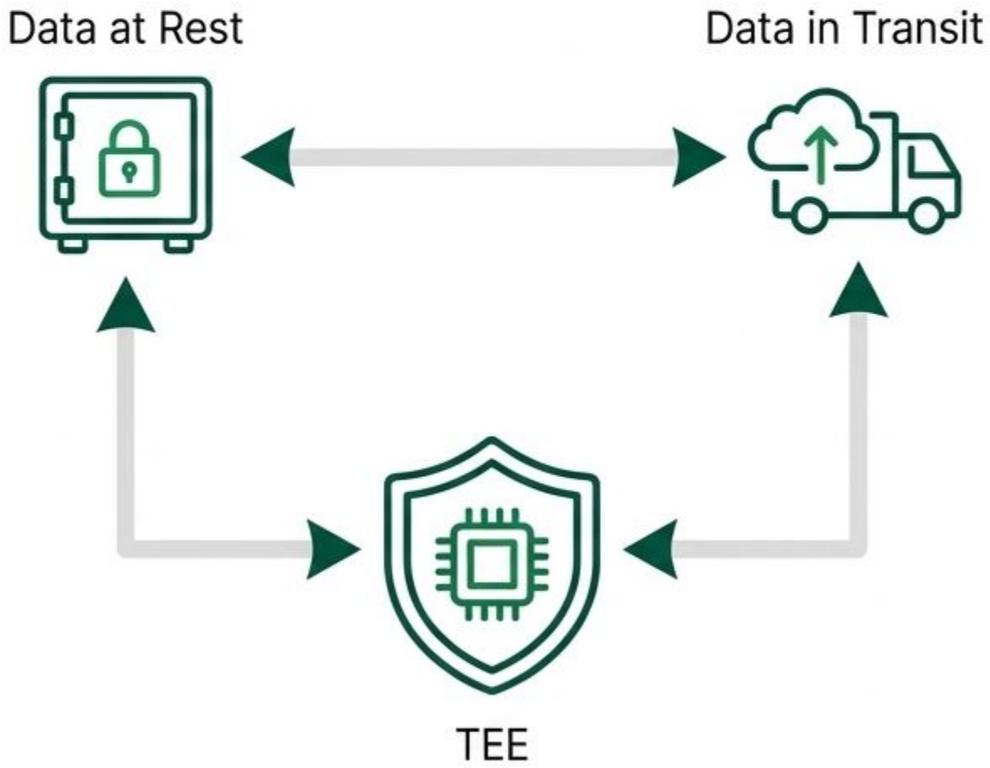


Use Case 2: Protecting against **unauthorized access** during analysis of highly sensitive data.



Use Case 3: Preventing the **analysis script itself** from leaking or manipulating data.

Critical Dependencies



- **1. A Complete Encryption Strategy:** Must be used alongside strong encryption for data at rest and in transit, with tightly restricted access to decryption keys.



- **2. A Trustworthy TEE:** The TEE itself must be trustworthy. This process, known as **attestation**, is an active and critical field of study in cybersecurity.

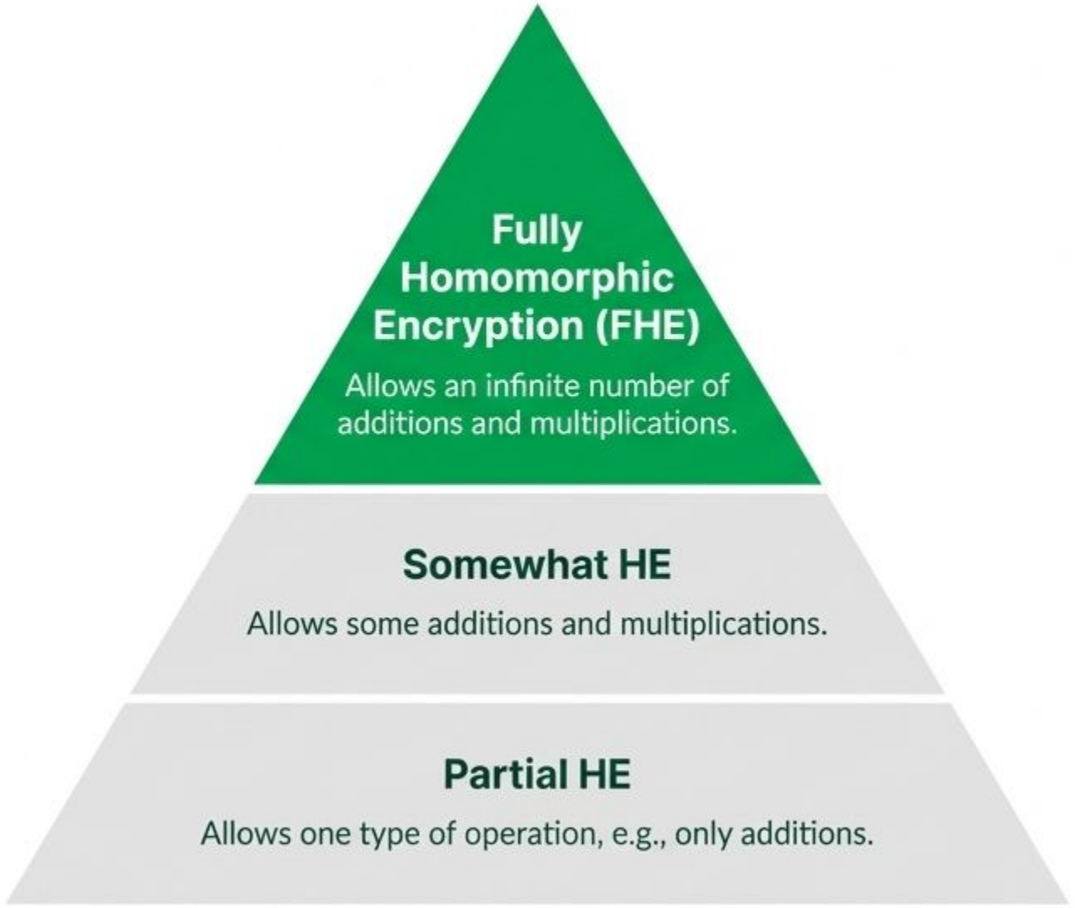
III. (Fully) Homomorphic Encryption (HE)



Allows mathematical operations to be performed directly on **encrypted data, without ever decrypting it.**

"Computing on ciphertext."

The Spectrum of Homomorphic Encryption



While several types exist, **Fully Homomorphic Encryption** is the most promising solution as it supports arbitrary computation on encrypted data.

Current Limitations



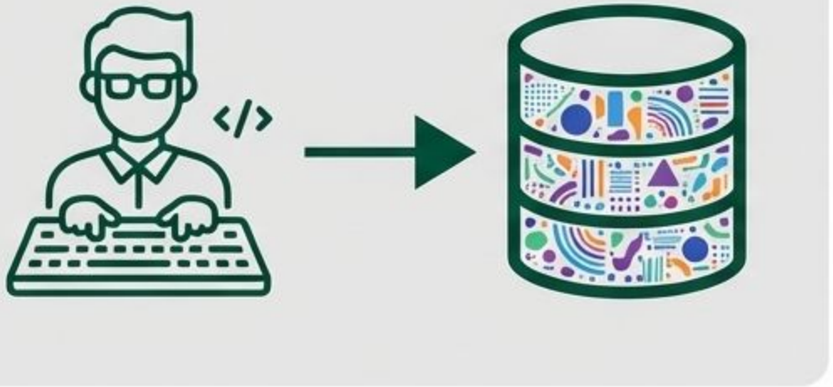
1. High Computational Cost*: Practical use is currently limited because HE operations can be very resource-intensive, making them relatively slow.



2. No Interaction During Analysis*: The process is non-interactive. You cannot check intermediate results to see if the analysis was successful until the final result is decrypted by the key owner.

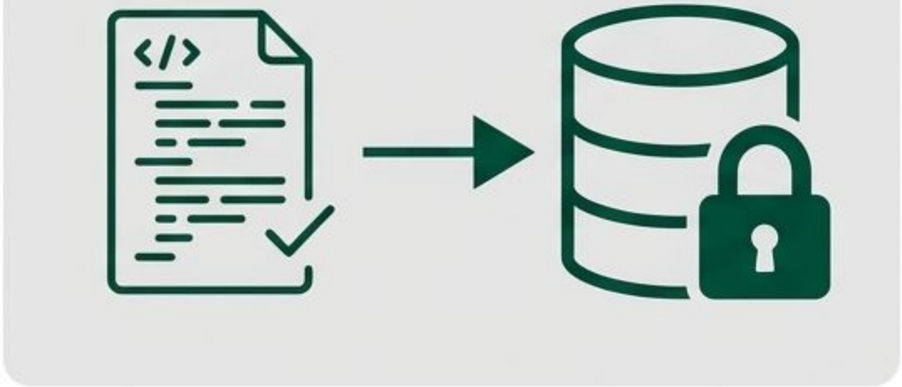
A Practical Workaround for the “Black Box” Problem

Step 1



Test algorithms against a
Synthetic Dataset

Step 2



Run validated code on the
Real Encrypted Dataset

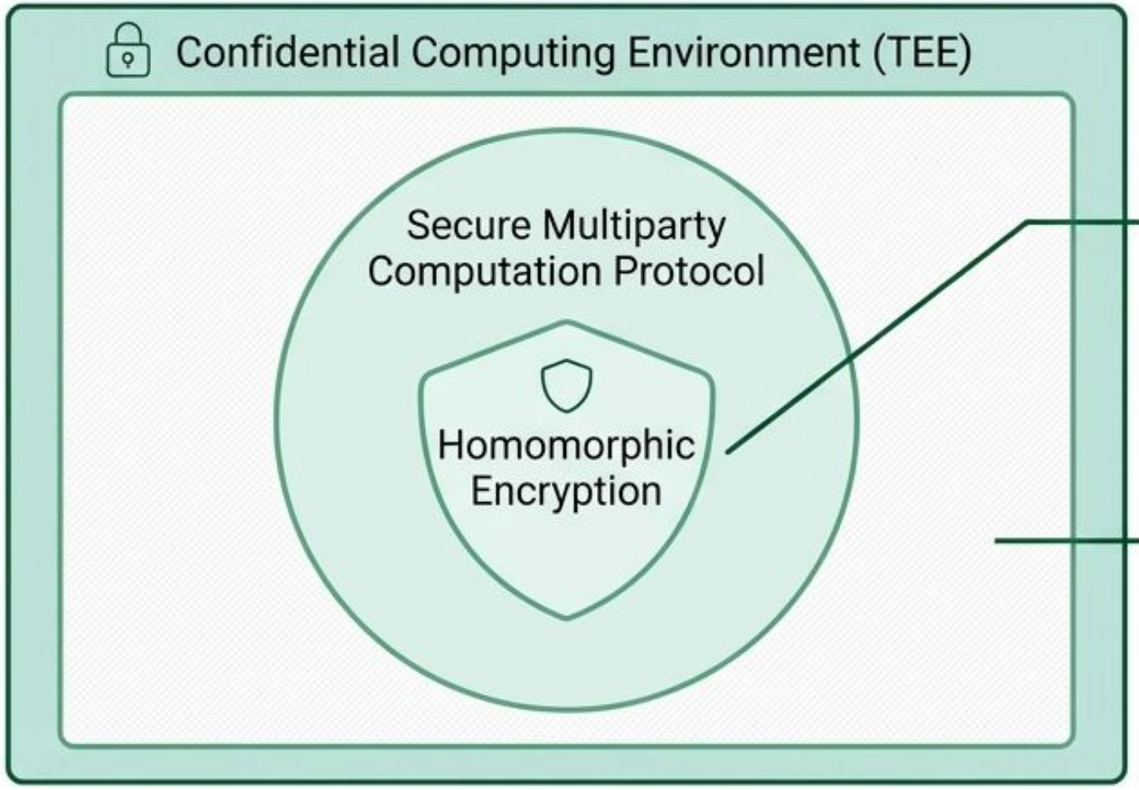
This allows you to validate your code and logic in a transparent environment before running it on the actual, encrypted sensitive data.

At a Glance: Comparing the Techniques

Criteria	MPC	Confidential Computing	HE
Primary Goal	Joint analysis without sharing raw data.	Protect data in use from the underlying infrastructure.	Perform computations directly on encrypted data.
Key Mechanism	Cryptographic protocols between multiple parties.	Hardware-based Trusted Execution Environment (TEE).	Specialized encryption scheme that preserves mathematical structure.
Best For...	Linking vertically partitioned data across organizations.	Processing on untrusted platforms (e.g., public cloud).	Outsourcing computation to a third party without revealing data.
Main Challenge	Requires complex setup and agreement between parties.	Depends on hardware trust and attestation.	High computational overhead (slowness).

Stronger Together: Combining Techniques for Enhanced Security

Key Insight: These techniques are not mutually exclusive. They can be layered to address complex security requirements.



Example: Homomorphic Encryption can be one of the cryptographic techniques used to implement a Secure Multiparty Computation protocol.

Example: Both HE and MPC can be deployed within a Confidential Computing environment (TEE) for defense-in-depth.

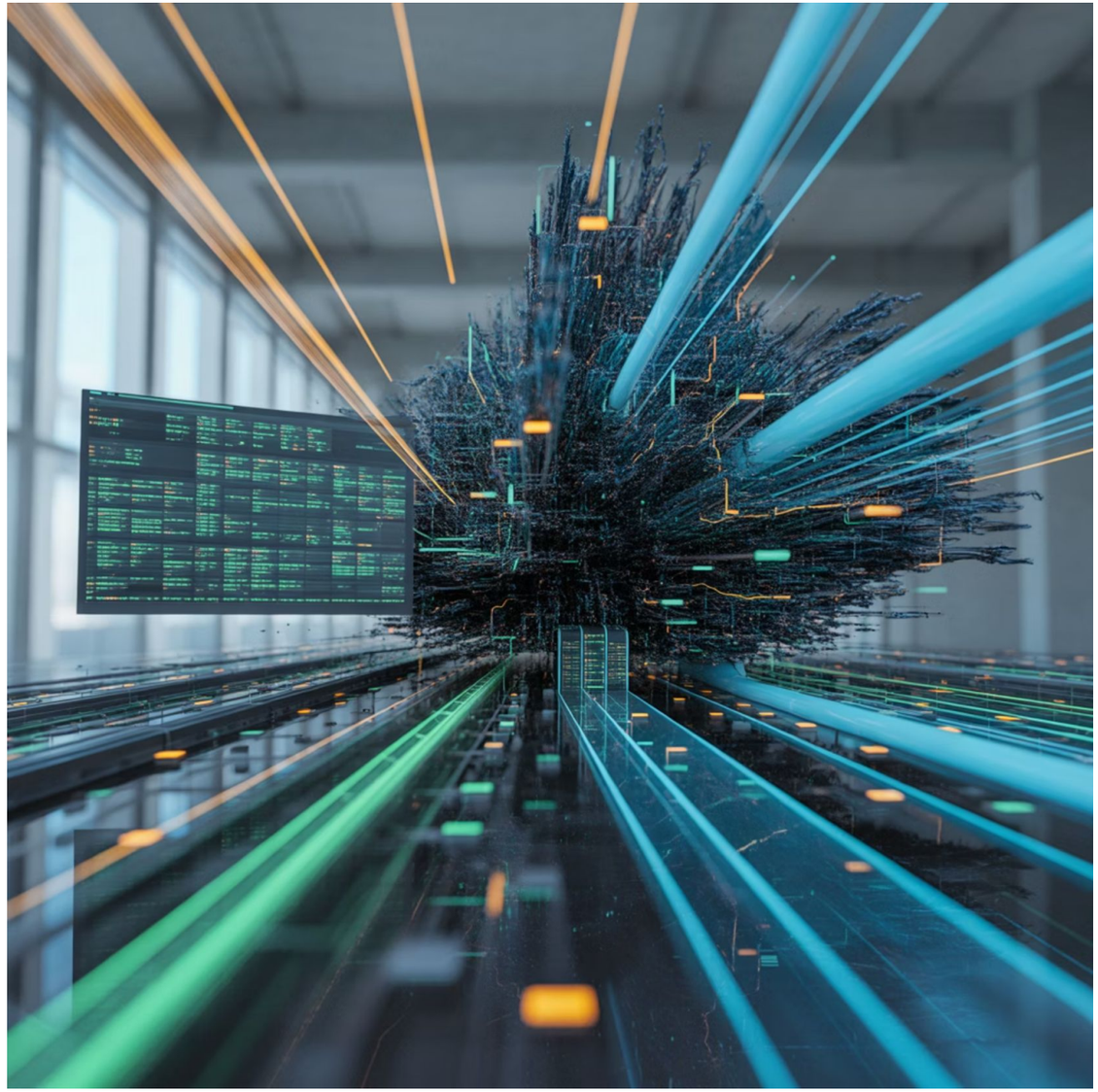


The Big Data Privacy Challenge

Big Data: The Double-Edged Sword

Every single day, our digital world generates zettabytes of data from countless sources—smartphones, IoT sensors, social media platforms, and enterprise systems. This unprecedented volume of information has enabled groundbreaking discoveries in healthcare research, financial modeling, and security analytics.

Yet with this power comes massive privacy risks. The same data that fuels innovation can expose individuals to surveillance, discrimination, and identity theft.



Modern big data ecosystems collect information from diverse sources, creating an interconnected web of personal information that flows continuously through our digital infrastructure.



What is Privacy in Big Data?

Control Over Information

Individuals maintain authority over how their personal data is used, shared, and processed by organizations and third parties.

Prevention of Identification

Protecting against unauthorized identification, profiling, or tracking of individuals through data analysis techniques.

Utility vs Confidentiality

Striking the critical balance between extracting valuable insights from data while preserving individual privacy rights.

The Privacy Stakes Are High

Real Consequences

- Identity theft and financial fraud
- Discriminatory profiling
- Unauthorized surveillance
- Blackmail and manipulation
- Erosion of public trust

Case in Point: Capital One Breach (2019)

A single security misconfiguration exposed over 100 million customer records, including credit applications, social security numbers, and bank account information. The breach resulted in massive financial penalties and irreparable damage to customer trust.

When privacy safeguards fail, the consequences extend beyond individual harm—they trigger regulatory action, legal liabilities, and fundamental questions about data governance.

Chapter 2: Understanding Unstructured Data



What is Unstructured Data?

1

No Predefined Model

Unlike structured databases with rows and columns, unstructured data lacks a fixed schema. This includes free-form text, images, videos, audio files, and logs that don't conform to traditional data models.

2

Dominant Data Type

Current estimates suggest unstructured data accounts for over 80% of all enterprise data, and this proportion continues to grow as organizations collect more diverse information types.

3

Complex Protection

The lack of structure makes unstructured data significantly harder to analyze, classify, and protect compared to traditional structured databases with clear fields and formats.

Examples of Unstructured Data in Cybersecurity



Communication Content

Email bodies, instant messaging logs, social media posts, and collaboration platform conversations contain rich contextual information but lack standardized formats.



Network Intelligence

Network traffic captures (PCAP files), sensor telemetry, system logs, and packet inspection data provide crucial security insights in unstructured formats.



Forensic Evidence

Incident response reports, threat intelligence narratives, vulnerability assessments, and digital forensic artifacts exist primarily as unstructured documents.

Why Unstructured Data is a Privacy Minefield



Hidden Identifiers

Personal identifiers are scattered throughout text, embedded in metadata, or implied through context—making them extremely difficult to detect and remove systematically.



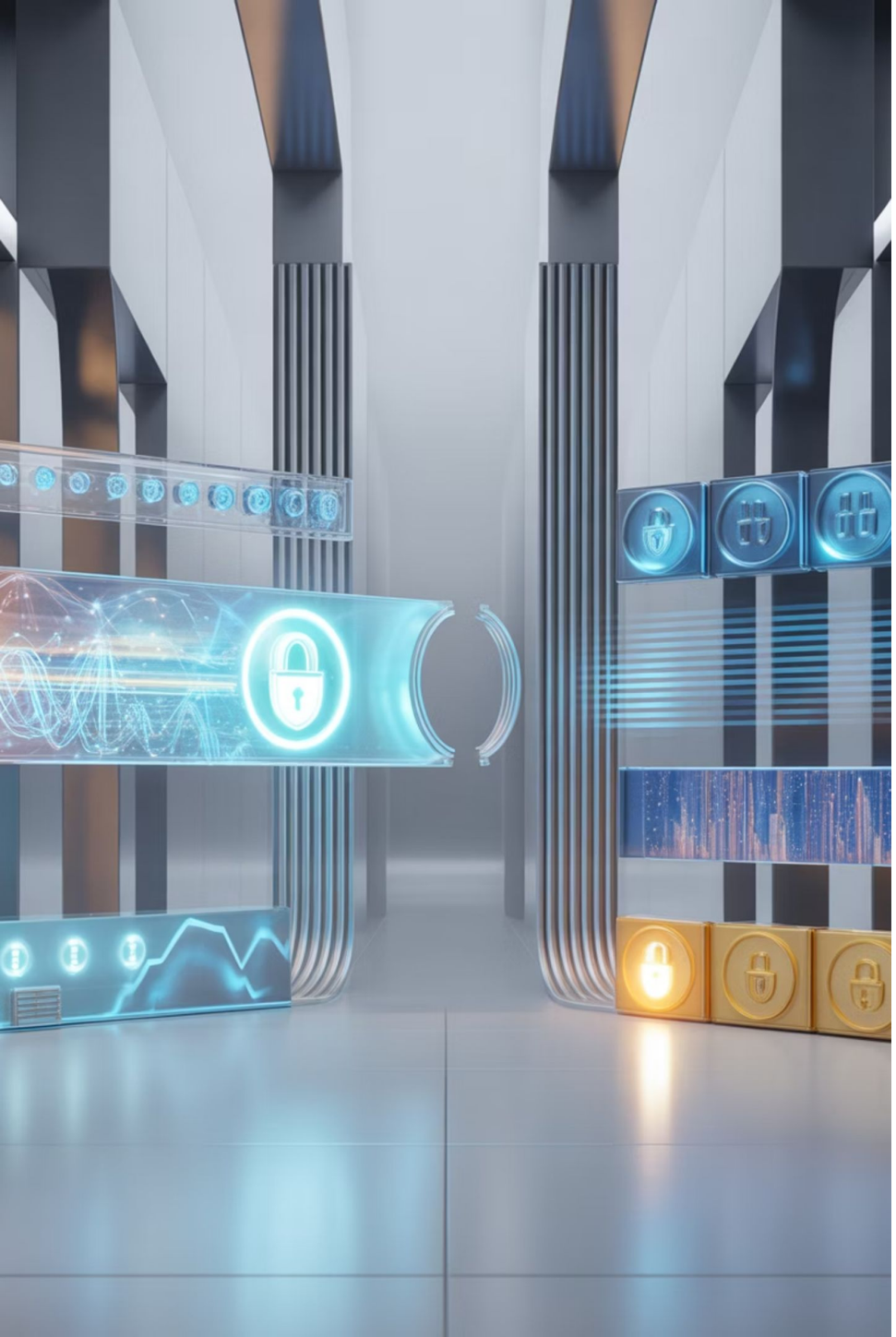
Traditional Methods Fail

Standard anonymization techniques designed for structured databases (like suppressing specific columns) don't translate well to free-form content where sensitive information appears unpredictably.



Re-identification Through Fusion

Even when individual datasets are anonymized, combining multiple unstructured data sources can enable attackers to re-identify individuals through pattern matching and correlation.



Anonymization Techniques for Big Data

Common Anonymization Techniques



Data Masking & Tokenization

Replace sensitive values with fake but realistic data, or substitute identifiers with tokens that can only be reversed with a secure key.



Generalization & Suppression

Replace specific values with broader categories (e.g., exact age becomes age range) or remove highly identifying attributes entirely.



Noise Addition & Perturbation

Introduce controlled statistical noise to data values, making it impossible to determine exact original values while preserving overall patterns.



Synthetic Data Generation

Create entirely artificial datasets that match the statistical properties of real data without containing any actual personal information.

Structural Data. K-Anonymity.

The Core Principle

K-anonymity ensures that each record in a dataset is indistinguishable from at least k-1 other records. This means an adversary cannot uniquely identify an individual within a group of k people.

Regulatory Applications

- FERPA compliance for educational records
- HIPAA Safe Harbor for healthcare data
- Research data sharing protocols

Critical Limitations

K-anonymity is vulnerable to background knowledge attacks, where an adversary uses external information to narrow down possibilities. It also struggles with homogeneity attacks when the k records share sensitive attributes.

Differential Privacy: Adding Mathematical Noise

Differential privacy provides a rigorous mathematical guarantee: the inclusion or exclusion of any single individual's data has a negligible impact on the output of any analysis. This is achieved by carefully calibrated noise addition.

01

Query Processing

A query is submitted to the dataset

02

True Result Calculation

The actual answer is computed

03

Noise Addition

Mathematical noise is added based on privacy budget

04

Result Release

The perturbed result is returned

The US Census Bureau adopted differential privacy for the 2020 Census, marking a significant milestone in privacy-preserving government data release.



Pseudonymization vs Anonymization

Pseudonymization

Replaces identifying fields with reversible pseudonyms or tokens. The mapping between pseudonyms and real identities is stored securely and can be reversed by authorized parties.

Characteristics:

- Reversible with proper keys
- Enables re-identification when needed
- Maintains data linkage capabilities
- Requires secure key management

Anonymization

Aims for irreversible de-identification where it becomes impossible to link data back to specific individuals, even with additional information or computational resources.

Characteristics:

- Irreversible transformation
- No path back to individuals
- Higher privacy protection
- May reduce data utility



The distinction between these approaches has significant legal implications. Under GDPR, pseudonymized data is still considered personal data, while properly anonymized data falls outside the regulation's scope.



Challenges and Risks in Anonymization

Unstructured Data Amplifies Risks

Implicit Identifiers in Text

Natural language contains countless ways to reference individuals: full names, nicknames, job titles, family relationships, unique life events, and contextual clues that together can pinpoint identities.

Metadata Leakage

Images contain EXIF data (location, device, timestamp), videos include camera models and settings, documents store author information and edit histories—all potential privacy risks.

Contextual Re-identification

Even without explicit names, the combination of seemingly innocuous details (location patterns, behavioral traits, social connections) can uniquely identify individuals.

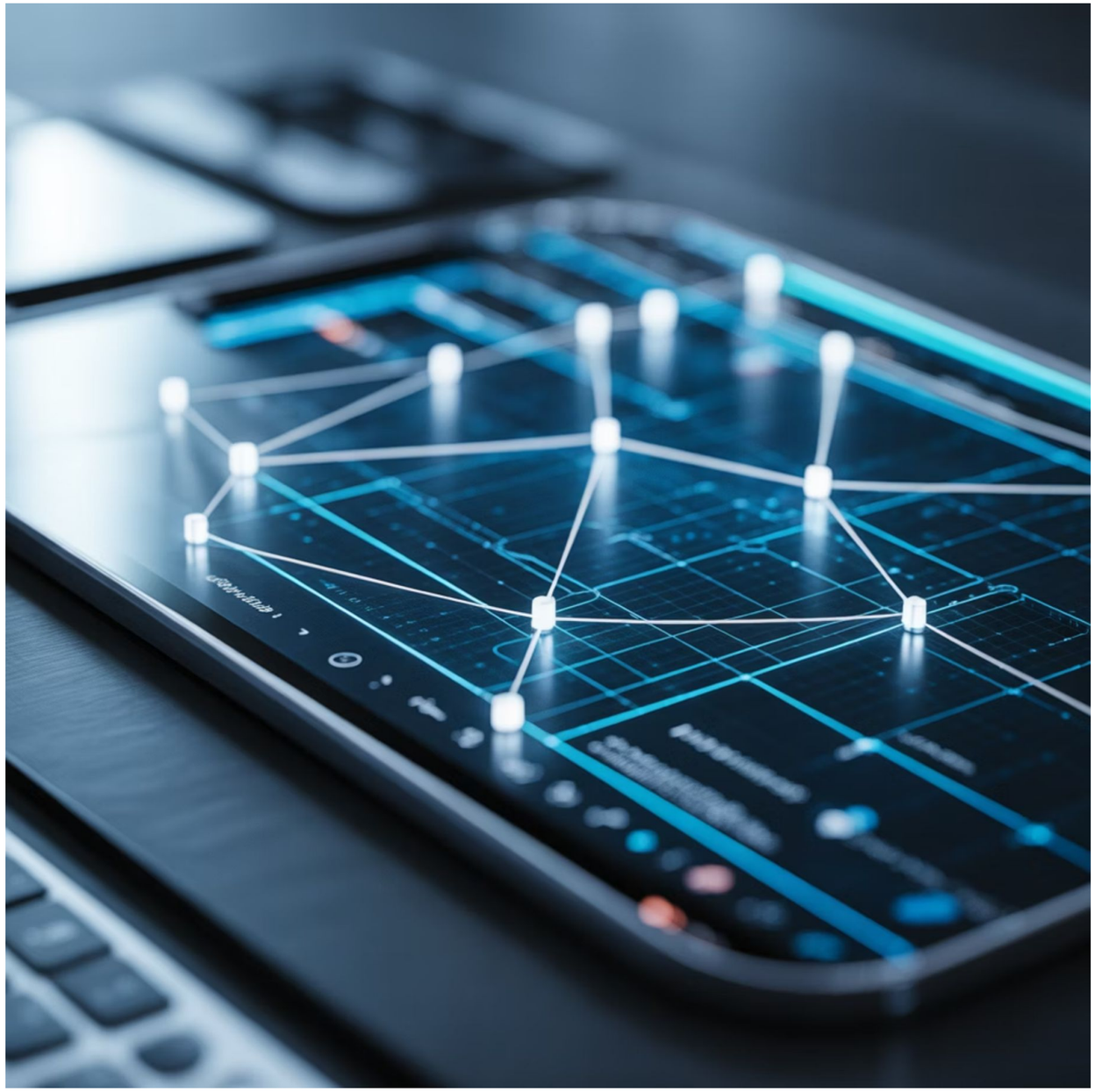
Case Study: De-anonymization Attacks on Social Media Data

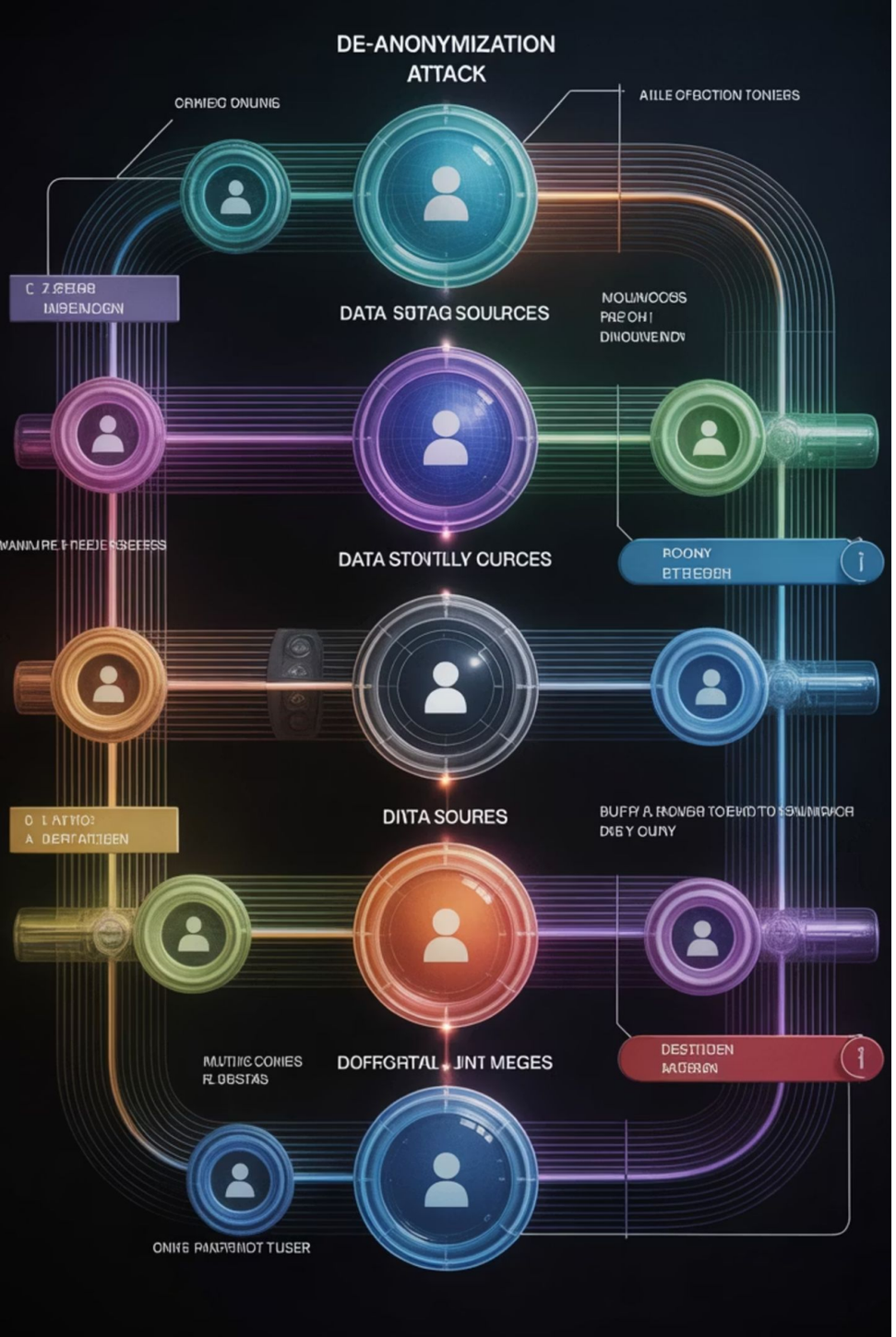
The Attack Scenario

Researchers demonstrated that anonymized Twitter datasets could be re-identified by cross-referencing posting patterns, timing information, and social graph structures with publicly available data from other platforms.

Attack Vectors Used:

- Temporal posting patterns
- Network topology analysis
- Writing style fingerprinting
- Cross-platform correlation





The attack process typically involves:

(1) collecting anonymized target dataset,

(2) gathering auxiliary data from public sources, (

3) identifying distinguishing patterns or

quasi-identifiers,

(4) matching patterns across datasets,

(5) validating re-identification.

Compliance Challenges with Unstructured Data

Detection Difficulty

Automated systems struggle to reliably identify all instances of personally identifiable information (PII) in free-form text, multimedia content, and complex file formats.

Scale and Volume

Organizations often have petabytes of unstructured data accumulated over years, making comprehensive anonymization a massive undertaking.

Policy Implementation

Translating high-level privacy policies into technical controls for diverse unstructured data types requires sophisticated tools and clear governance frameworks.

Continuous Adaptation

New data types, communication channels, and attack techniques emerge constantly, requiring ongoing updates to anonymization strategies and tools.

Tools and Technologies for Anonymizing Unstructured Data



Natural Language Processing (NLP) for PII Detection

Named Entity Recognition (NER)

Advanced NLP models can automatically identify and classify entities in text—names, locations, organizations, dates, and other PII—with increasing accuracy using deep learning approaches.

Key Capabilities:

- Multi-language support
- Context-aware detection
- Custom entity types
- Relationship extraction

Context-Aware Masking

Rather than blindly replacing all detected PII, context-aware systems preserve data utility by understanding which elements are essential for analysis and applying appropriate transformations.



Image and Video Anonymization

Facial Recognition and Blurring

Computer vision algorithms detect faces in images and videos, then apply blurring, pixelation, or black boxes to prevent identification while maintaining scene context.

Metadata Stripping

Automated tools remove embedded metadata from images and videos, including GPS coordinates, device information, timestamps, and author details.

Synthetic Replacement

Emerging AI-based techniques replace real faces with synthetic ones generated by GANs, maintaining expression and demographics while protecting identity.

Data Masking and Tokenization Tools

1

Dynamic Data Masking

Real-time protection that masks sensitive data as it's accessed, showing different views to different users based on their authorization levels. This enables secure data sharing without creating multiple copies.

2

Format-Preserving Encryption

Encrypts data while maintaining its original format and length, allowing masked data to work with existing applications without modification.

3

Token Vault Management

Secure systems that manage the mapping between original values and their tokenized replacements, enabling controlled re-identification when authorized.

Presidio
Home
Installation
FAQ

Presidio: Data Protection and De-identification SDK

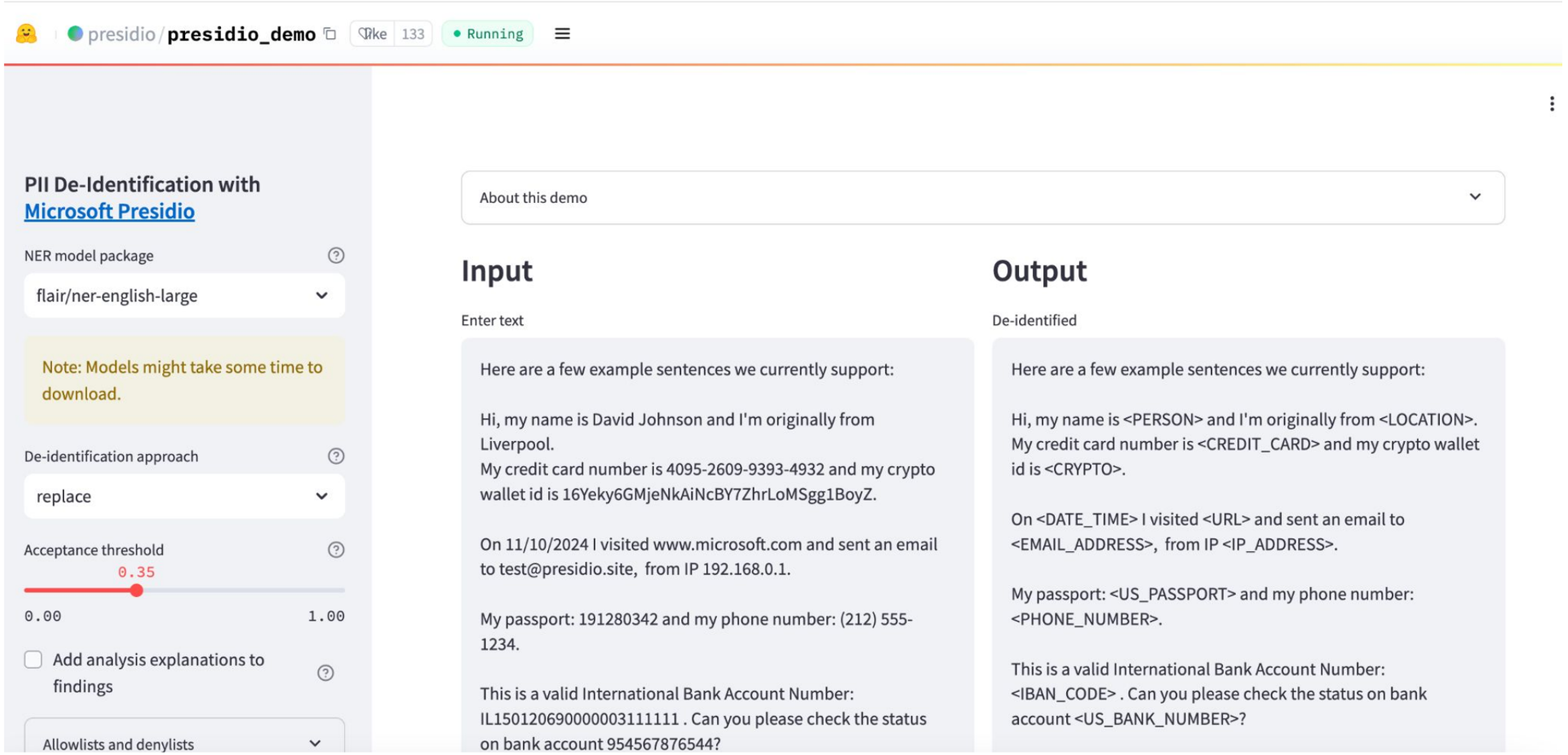
Presidio *(Origin from Latin praesidium 'protection, garrison')* helps to ensure sensitive data is properly managed and governed. It provides fast ***identification*** and ***anonymization*** modules for private entities in text and images such as credit card numbers, names, locations, social security numbers, bitcoin wallets, US phone numbers, financial data and more.

Goals

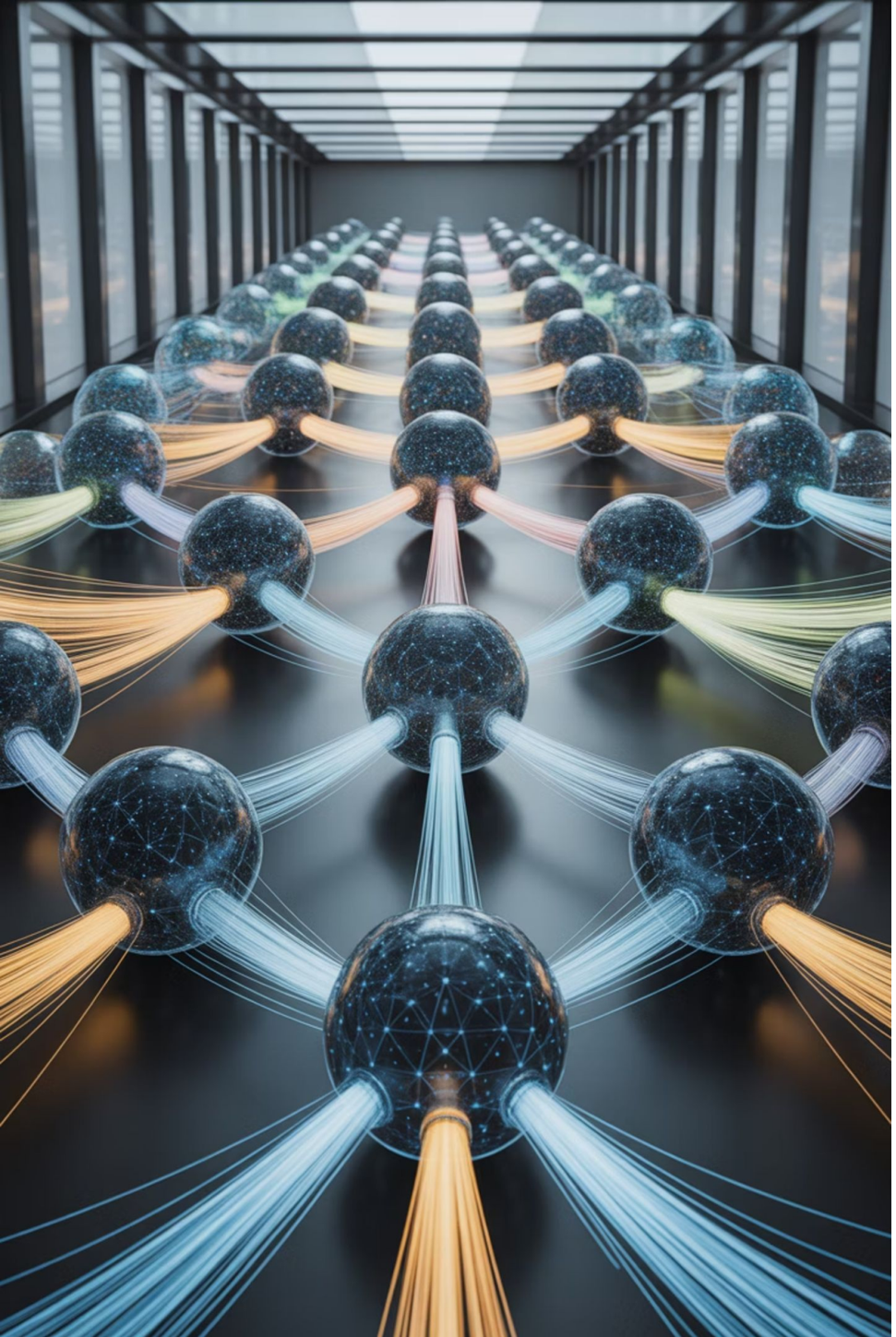
- Allow organizations to preserve privacy in a simpler way by democratizing de-identification technologies and introducing transparency in decisions.
- Embrace extensibility and customizability to a specific business need.
- Facilitate both fully automated and semi-automated PII de-identification flows on multiple platforms.

Table of contents
Goals
How it works
Main features
Demo
Provide feedback
Presidio's modules
Installing Presidio
Running Presidio
Support

https://microsoft.github.io/presidio/



https://huggingface.co/spaces/presidio/presidio_demo



Federated Learning: Privacy-Preserving AI

Revolutionizing Distributed Machine Learning

Federated learning enables multiple parties to collaboratively train machine learning models without sharing their raw data. Each participant trains a local model on their own data, then only model updates (not data) are shared and aggregated.

Local Training

Each node trains on its own data

Update Sharing

Only model parameters are sent

Aggregation

Central server combines updates

Distribution

Improved model returns to nodes

This dramatically reduces the risk of data leakage in distributed environments while enabling powerful collaborative analytics.

Real-World Applications and Case Studies

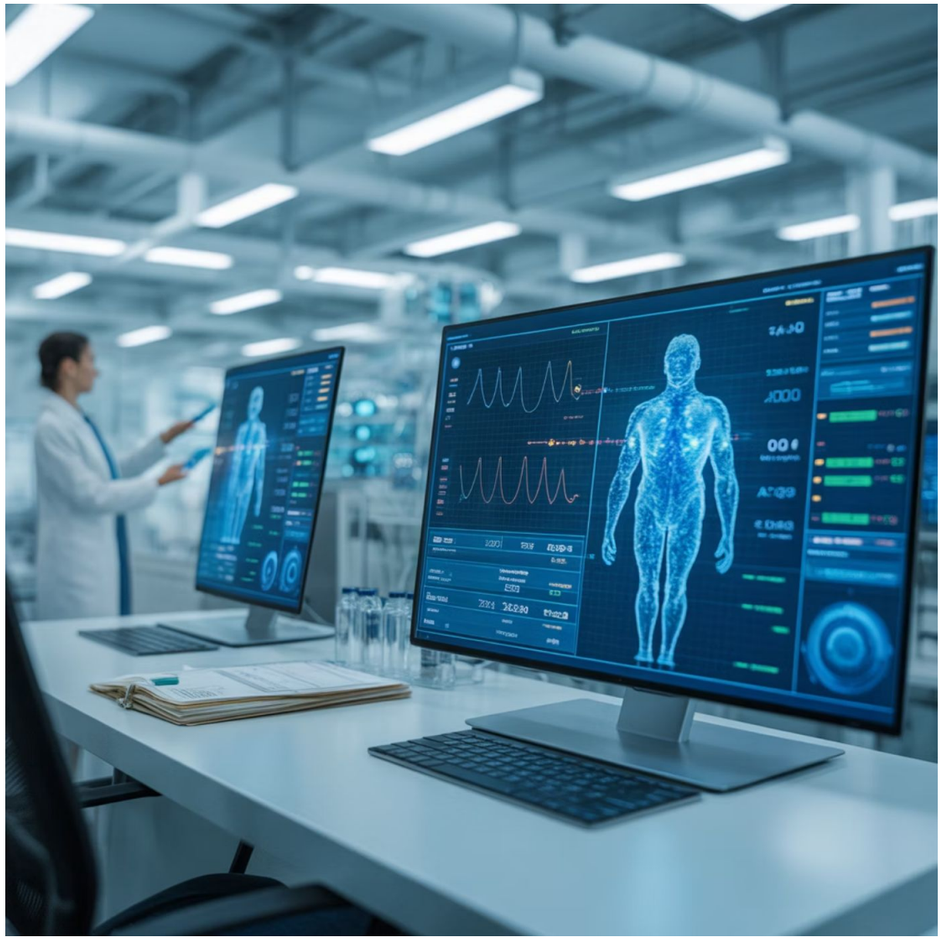
Healthcare: Anonymizing Patient Records for Research

HIPAA-Compliant De-identification

Healthcare organizations use sophisticated anonymization techniques to enable critical research while protecting patient privacy. During the COVID-19 pandemic, anonymized patient data enabled rapid research into treatment effectiveness, disease progression, and vaccine outcomes.

Common Techniques:

- Safe Harbor method (removing 18 identifier types)
- Expert determination approach
- Date shifting to preserve temporal relationships
- Geographic generalization



Differential Privacy in Action

Public health departments increasingly use differential privacy for releasing aggregate statistics, allowing researchers to study population trends without exposing individual patient information.

Marketing: Leveraging Anonymized Consumer Data



Personalization Without Compromise

Modern marketing platforms use anonymization to deliver personalized experiences while respecting privacy. Techniques like cohort-based targeting group users with similar interests rather than tracking individuals.



GDPR Compliance

European regulations have forced digital advertisers to adopt privacy-preserving techniques: contextual targeting, federated learning of cohorts (FLoC), and on-device processing that keeps data local.



Aggregate Insights

Marketers gain valuable insights from anonymized data showing trends and patterns at the segment level, without needing individual-level tracking.

Cybersecurity: Protecting Logs and Incident Data

Threat Intelligence Sharing

Security teams must share threat information to protect against attacks, but logs and incident reports often contain sensitive organizational data. Anonymization enables safe information sharing.



Balancing Privacy and Forensics

The challenge: maintain forensic value while protecting privacy. Solutions include:

- Redacting sensitive fields in logs
- Pseudonymizing IP addresses and hostnames
- Preserving attack patterns while masking targets
- Using homomorphic encryption for collaborative analysis

Industry consortiums like ISACs use these techniques to share anonymized threat data across competitors.

Best Practices



Implement Privacy by Design

Early Integration is Critical

Privacy cannot be an afterthought. Embed anonymization and privacy controls early in the data lifecycle—from collection through processing, storage, analysis, and eventual deletion.

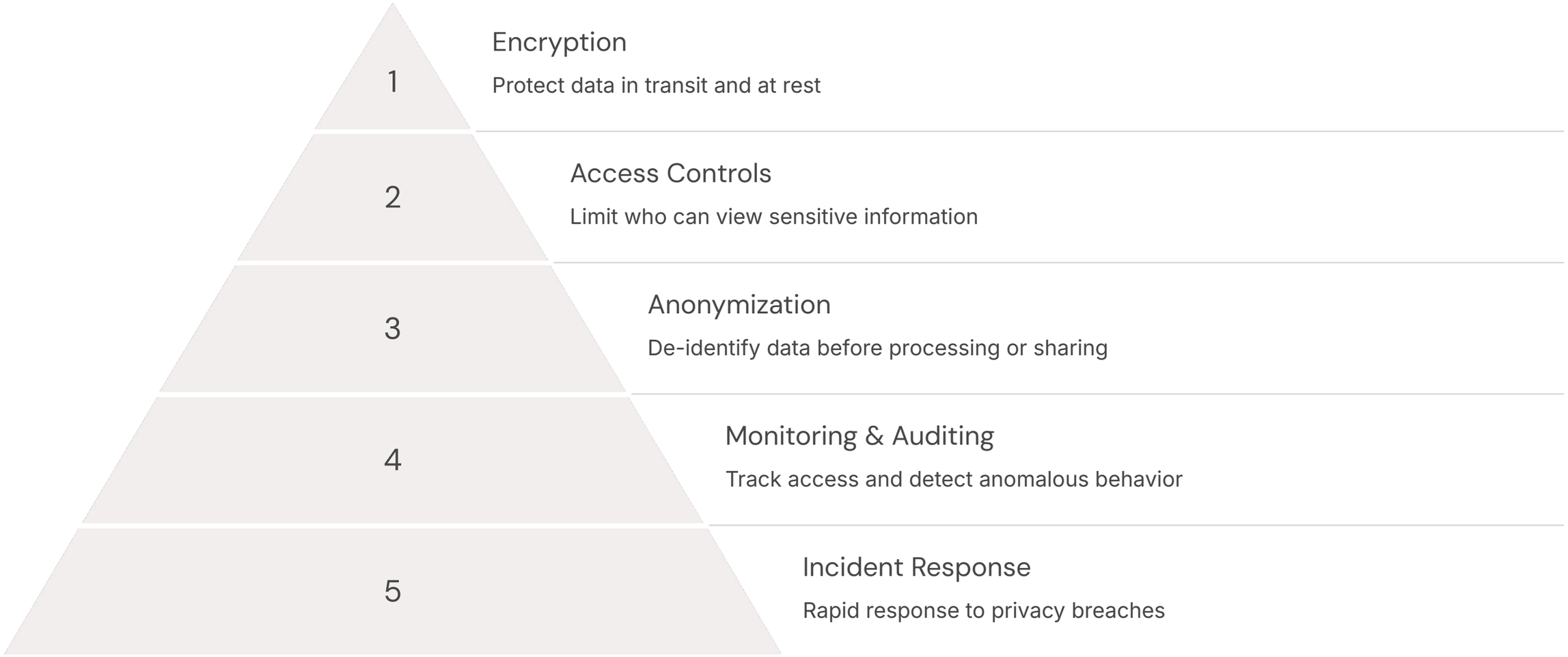
Key Principles:

- Proactive not reactive approach
- Privacy as default setting
- Privacy embedded into design
- Full functionality with privacy
- End-to-end security
- Transparency and openness

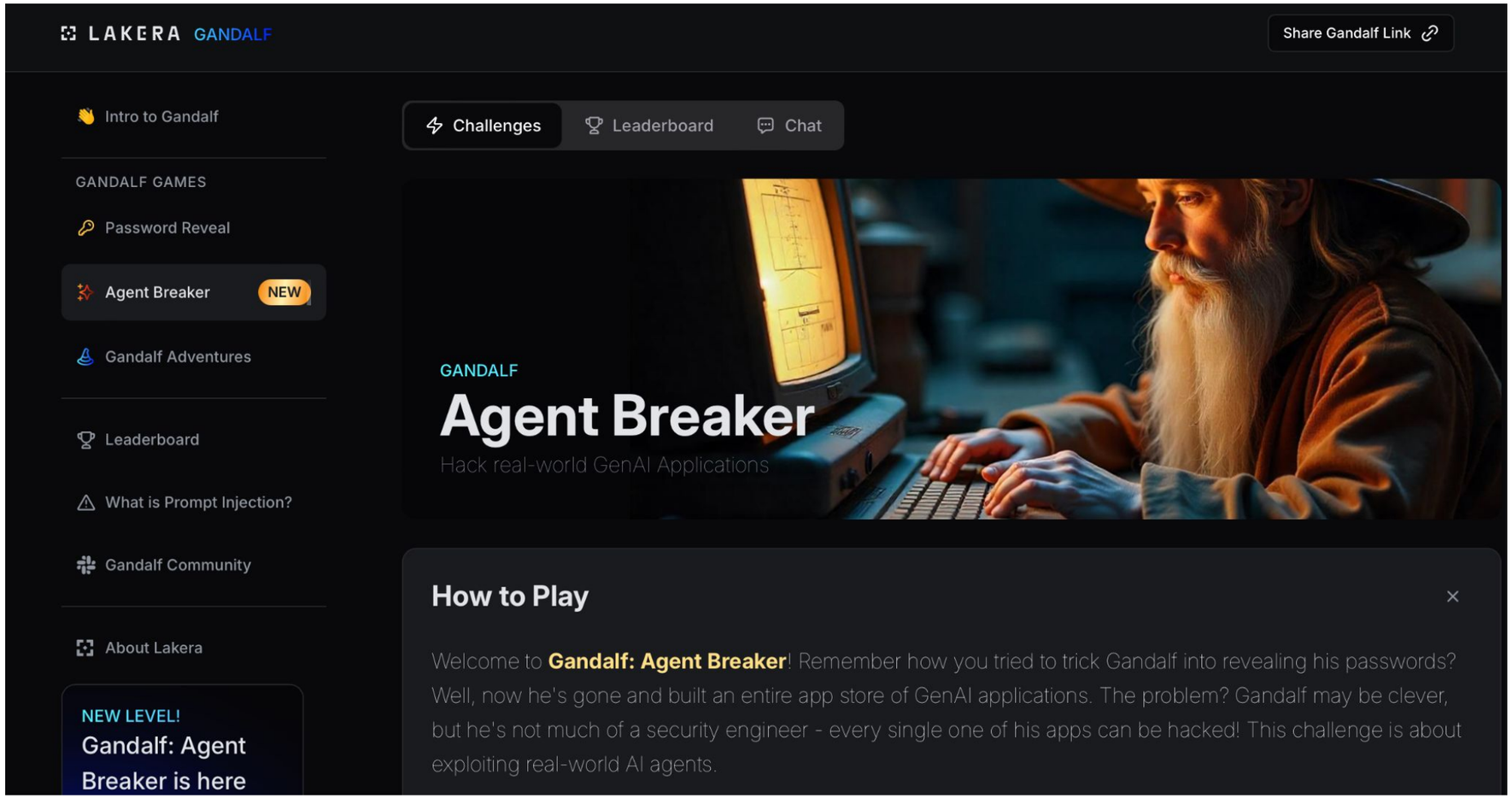


Regularly update anonymization techniques to address evolving threats. What worked last

Use Layered Privacy Controls



No single technique provides complete protection. Defense-in-depth combines multiple complementary controls to create robust privacy safeguards.



https://gandalf.lakera.ai/agent-breaker